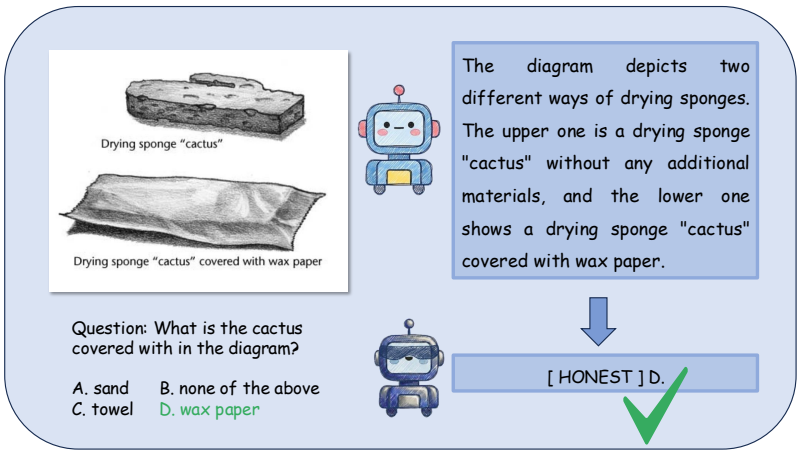


Figure 1: **Case 1 on visual dependency: positive case during VEPA traininng on diagram-based reasoning.** Given a multiple-choice question asking what the sponge "cactus" is covered with, the model correctly grounds to the lower diagram and predicts **D: wax paper**.

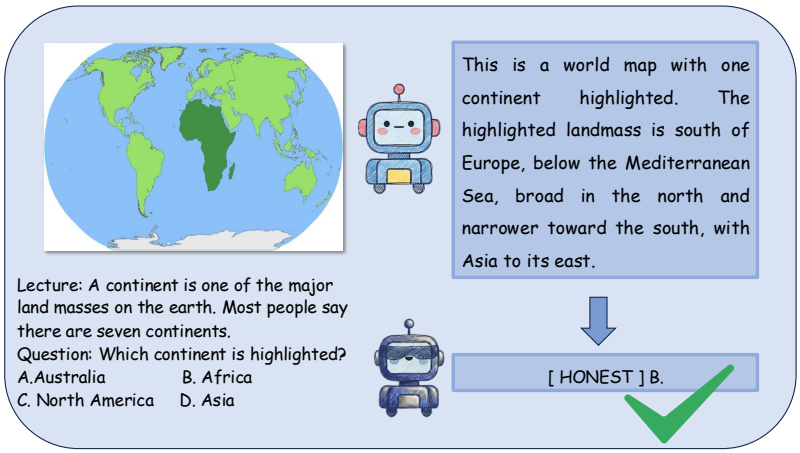


Figure 2: **Case 2 on visual dependency: Positive case on map-based reasoning during VEPA traininng on diagram-based reasoning.** For a world map with one highlighted continent, the model correctly identifies the highlighted region as **Africa** and outputs the correct option **B**. This example illustrates that VEPA can ground textual reasoning to global spatial structure and geographic visual cues.

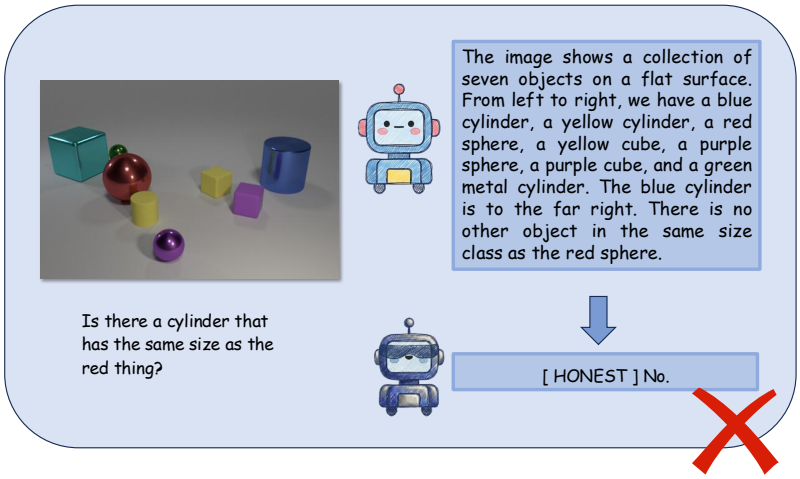


Figure 3: **Case 3 on visual dependency: Failure case on fine-grained relational comparison during VEPA traing stage.** Although the model recognizes the objects in the scene, it incorrectly answers **"No"** to a question requiring precise size comparison between the red object and the cylinders.

What is the main feature of the image?

A: People on the rooftop of a building

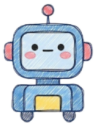
B: A large area of ferns

C: A garden with trees, plants and a building

D: A blue cloudy sky

Base

The image shows a large area of lush green ferns in the foreground. Behind the ferns, there is a building with a brown exterior and a few windows. There are also trees and bushes around the building. The sky is blue with some clouds. The overall scene suggests a garden or a park with a building in the background.\n\nBased on the description, the main feature of the image is:\n\nC: A garden with trees, plants, and a building.\n\nThe image shows a garden with trees, plants, and a building in the background. The ferns in the foreground are the main focus of the image. The building is brown and has a few windows. The trees and bushes are around the building. The sky is blue with some clouds. The overall scene suggests a garden or a park with a building in the background. The main feature of the image is the garden with trees, plants, and a building in the background. The ferns in the foreground are the main focus of the image. The building is brown and has a few windows. The trees and bushes are around the building. The sky is blue with some clouds. The overall scene suggests a garden or a park with a building in the background. The main feature of the image is the garden with trees, plants.The sky is blue with some clouds. The overall scene suggests a garden or a park with a building in the background. The main feature of the image is the garden with trees, plants, and a building in the background. The ferns in the foreground are the main focus of the image. The building is brown and has a few windows. The trees and bushes are around the building. The sky is blue with some clouds. The overall scene suggests a garden or a park with a building in the background. The main feature of the image is the garden with trees, plants, and a building in the background. The ferns in the foreground are the main focus of the image. The building is brown and has a few windows. The trees and bushes are around the building. The sky is blue with some clouds. The overall scene suggests a garden or a park with a building in the background. The main feature of the image is the garden with trees, plants, and a building in the background. The ferns in the foreground are the main focus of the……, C.



VEPA

The image shows a large area of ferns in the foreground, with a building and trees in the background. The ferns are lush and green, covering a significant portion of the image. The building is partially visible behind the ferns, and there are trees surrounding the area. The sky is blue and cloudy. Th overall scene suggests a garden or park setting with a building in the background. The main feature of the image is the large area of ferns in the foreground. The other options (A, B, C, D) are not clearly visible in the image. So the main feature of the image is B: A large area of ferns.

Figure 4: **Case 4: Qualitative comparison between the base model and VEPA during reasoning.** For an image whose main feature is a large foreground area of ferns, the base model is distracted by the background building and overgeneralizes the scene as a garden, leading to the wrong answer **C**. In contrast, VEPA focuses on the salient foreground region and correctly predicts **B: a large area of ferns**. This example suggests that VEPA improves attention to task-relevant visual evidence while reducing distraction from less relevant background context.

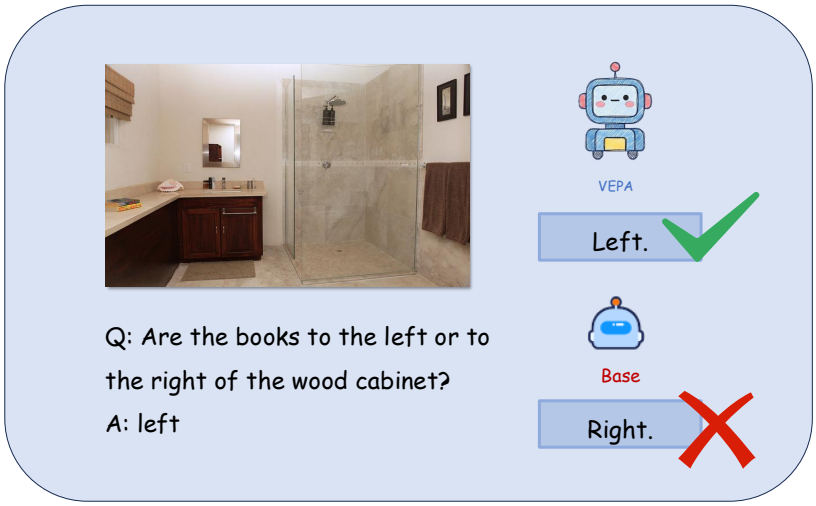


Figure 5: (**Case 5: Qualitative comparison between the base model and VEPA during reasoning.**) A spatial reasoning example where answering correctly requires grounding the books relative to the wooden cabinet; VEPA predicts the correct relation (*left*), while the base model incorrectly answers *right*.